SIMATS ENGINEERING
ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 20%

Scenario-Based Questions

1. Low Contrast Image
A grayscale image appears dull with very little difference between object and background intensities.
(a) Clearly interpret the scenario and explain the cause of low contrast.
(b) Identify all key issues affecting image visibility.
(c) Apply an appropriate enhancement technique to address the problem.
(d) Justify your choice with logical reasoning.
(e) Present your explanation in a clear and structured manner.

2. Uneven Illumination
An image captured under non-uniform lighting shows bright and dark patches.
(a) Interpret the scenario and explain the impact of uneven illumination.
(b) Identify key factors causing intensity variation.
(c) Apply a suitable enhancement method to normalize illumination.
(d) Justify your selected approach with reasoning.
(e) Provide a clear and well-structured explanation.

3. Salt-and-Pepper Noise
A scanned document contains salt-and-pepper noise affecting readability.
(a) Explain the scenario and characteristics of this noise.
(b) Identify key issues impacting image quality.
(c) Apply an appropriate filtering technique.
(d) Justify why this method is more suitable than alternatives.
(e) Present your response clearly and logically.

4. Blurred Edges
An image processed for noise removal appears overly smooth, causing edges to blur.
(a) Interpret the scenario and explain why edges are blurred.
(b) Identify key issues related to smoothing and edge loss.
(c) Apply an appropriate technique to restore edges.
(d) Justify your decision with clear reasoning.
(e) Structure your explanation clearly.

5. High-Frequency Noise
A satellite image contains fine-grained noise affecting feature visibility.
(a) Interpret the scenario and describe high-frequency noise.
(b) Identify the key issues affecting image clarity.
(c) Apply an appropriate frequency domain filtering method.
(d) Justify the choice of filter with reasoning.
(e) Present your answer in a clear and organized manner.

6. Object Boundary Detection
In a medical image, boundaries between tissues need to be clearly identified.
(a) Interpret the scenario and explain the importance of boundary detection.
(b) Identify key challenges in detecting boundaries.
(c) Apply a suitable edge detection method.
(d) Justify your selection with logical reasoning.
(e) Present your explanation clearly and systematically.

7. Simple Segmentation
An image contains objects clearly distinguishable from the background based on intensity.
(a) Interpret the scenario and explain segmentation requirements.
(b) Identify key features enabling separation.
(c) Apply a suitable segmentation technique.
(d) Justify your decision logically.
(e) Present your response clearly.

8. Complex Segmentation
An image contains multiple regions with overlapping intensity values.
(a) Interpret the scenario and explain segmentation challenges.
(b) Identify key issues such as intensity overlap.
(c) Apply an appropriate advanced segmentation method.
(d) Justify how the method handles complexity.
(e) Provide a structured and clear explanation.

9. Broken Edges
After applying edge detection, object boundaries appear discontinuous.
(a) Interpret the scenario and explain why edges are broken.
(b) Identify key issues affecting boundary continuity.
(c) Apply a suitable method to improve edge connectivity.
(d) Justify your approach logically.
(e) Present your explanation clearly.

10. Combined Requirement
An industrial inspection system requires noise removal while preserving important edges for defect detection.
(a) Interpret the scenario and define the combined requirement.
(b) Identify key issues involving noise and edge preservation.
(c) Apply an appropriate sequence of techniques.
(d) Justify the order and selection with strong reasoning.
(e) Present your response in a clear and structured format.

11. Low Brightness Image

An image captured in low light appears very dark with poor visibility.
(a) Interpret the scenario and explain the cause of low brightness.
(b) Identify key issues affecting image visibility.
(c) Apply an appropriate enhancement technique.
(d) Justify your choice with reasoning.
(e) Present your explanation clearly.

12. Overexposed Image

An image appears excessively bright, losing important details.
(a) Interpret the scenario and explain overexposure.
(b) Identify key issues affecting intensity representation.
(c) Apply a suitable correction technique.
(d) Justify your choice logically.
(e) Present your explanation in a structured manner.

13. Gaussian Noise

An image contains Gaussian noise affecting smooth regions.
(a) Interpret the scenario and describe Gaussian noise.
(b) Identify key issues affecting quality.
(c) Apply an appropriate smoothing filter.
(d) Justify your method over alternatives.
(e) Present your explanation clearly.

14. Speckle Noise

An image obtained from ultrasound imaging shows speckle noise.
(a) Interpret the scenario and explain speckle noise characteristics.
(b) Identify key issues affecting clarity.
(c) Apply a suitable filtering technique.
(d) Justify your approach logically.
(e) Present your explanation clearly.

15. Loss of Fine Details

An image loses fine details after processing.
(a) Interpret the scenario and explain detail loss.
(b) Identify key issues related to filtering.

(c) Apply a suitable enhancement technique.
(d) Justify your method with reasoning.
(e) Present your explanation clearly.

16. Edge Enhancement Requirement

An application requires highlighting edges for feature extraction.
(a) Interpret the scenario and explain the need for edge enhancement.
(b) Identify key issues in detecting edges.
(c) Apply an appropriate method.
(d) Justify your choice logically.
(e) Present your explanation clearly.

17. Background Noise Removal

An image contains background noise affecting segmentation accuracy.
(a) Interpret the scenario and explain noise impact.
(b) Identify key issues affecting segmentation.
(c) Apply a suitable preprocessing method.
(d) Justify your approach logically.
(e) Present your explanation clearly.

18. Threshold Selection Problem

An image requires segmentation, but selecting a threshold is difficult.
(a) Interpret the scenario and explain the challenge.
(b) Identify key issues in threshold selection.
(c) Apply a suitable thresholding method.
(d) Justify your approach logically.
(e) Present your explanation clearly.

19. Texture-Based Segmentation

An image contains regions distinguishable by texture rather than intensity.
(a) Interpret the scenario and explain segmentation challenges.
(b) Identify key issues with intensity-based methods.
(c) Apply an appropriate segmentation approach.
(d) Justify your method logically.
(e) Present your explanation clearly.

20. Multiple Object Detection

An image contains multiple objects with similar intensity values.
(a) Interpret the scenario and explain segmentation difficulty.
(b) Identify key issues in distinguishing objects.
(c) Apply a suitable segmentation technique.
(d) Justify your choice logically.
(e) Present your explanation clearly.

21. Edge Noise Problem

Edges detected are affected by noise.
(a) Interpret the scenario and explain the issue.
(b) Identify key challenges in edge detection.
(c) Apply a suitable solution.
(d) Justify your method logically.
(e) Present your explanation clearly.

22. Region Merging Requirement

An image contains fragmented regions that need merging.
(a) Interpret the scenario and explain the issue.
(b) Identify key challenges in region segmentation.
(c) Apply an appropriate technique.
(d) Justify your approach logically.
(e) Present your explanation clearly.

23. Over-Segmentation

An image is divided into too many regions after segmentation.
(a) Interpret the scenario and explain over-segmentation.
(b) Identify key issues affecting segmentation.
(c) Apply a suitable correction method.
(d) Justify your choice logically.
(e) Present your explanation clearly.

24. Under-Segmentation

Important regions are merged incorrectly during segmentation.
(a) Interpret the scenario and explain under-segmentation.
(b) Identify key issues affecting separation.
(c) Apply a suitable solution.
(d) Justify your approach logically.
(e) Present your explanation clearly.

25. Boundary Refinement

Detected boundaries are rough and inaccurate.
(a) Interpret the scenario and explain the issue.
(b) Identify key challenges in boundary detection.
(c) Apply a suitable refinement method.
(d) Justify your choice logically.
(e) Present your explanation clearly.

26. Frequency-Based Enhancement

An image requires enhancement using frequency domain methods.
(a) Interpret the scenario and explain frequency-based processing.

(b) Identify key issues in spatial domain methods.
(c) Apply an appropriate frequency domain technique.
(d) Justify your choice logically.
(e) Present your explanation clearly.

27. Mixed Noise Scenario

An image contains both Gaussian and impulse noise.
(a) Interpret the scenario and explain challenges.
(b) Identify key issues affecting filtering.
(c) Apply a suitable combined approach.
(d) Justify your method logically.
(e) Present your explanation clearly.

28. Edge Preservation Filtering

A filtering method is required to remove noise but preserve edges.
(a) Interpret the scenario and explain the requirement.
(b) Identify key issues in filtering.
(c) Apply a suitable technique.
(d) Justify your approach logically.
(e) Present your explanation clearly.

29. Region Growing Failure

Region growing fails due to incorrect seed selection.
(a) Interpret the scenario and explain the issue.
(b) Identify key challenges in region growing.
(c) Apply a suitable correction.
(d) Justify your method logically.
(e) Present your explanation clearly.

30. Industrial Inspection Enhancement

An industrial system requires detecting defects in noisy images.
(a) Interpret the scenario and define requirements.
(b) Identify key issues in detection.
(c) Apply an appropriate enhancement and segmentation pipeline.
(d) Justify your sequence logically.
(e) Present your explanation clearly.

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

		presented			